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Heart Failure Clinical Records

Donated on 2/4/2020

This dataset contains the medical records of 299 patients who had heart failure, collected during their follow-up period, where each patient profile has 13 clinical features.

Dataset Characteristics

Multivariate

Subject Area

Health and Medicine

Associated Tasks

Classification, Regression, Clustering

Feature Type

Integer, Real

Instances

299

Features

12



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CITE

1 citations

120835 views

DOI

10.24432/C5Z89R

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Dataset Information

Additional Information

A detailed description of the dataset can be found in the Dataset section of the following paper:

Davide Chicco, Giuseppe Jurman: "Machine learning can predict survival of patients with heart failure from seru...

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Has Missing Values?

No

Introductory Paper

[Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone](#)

By D. Chicco, Giuseppe Jurman. 2020

Published in BMC Medical Informatics and Decision Making

Variables Table

Variable Name	Role	Type	Demographic	Description	Units	Missing V
age	Feature	Integer	Age	age of the patient	years	no
anaemia	Feature	Binary		decrease of red blood cells or hemoglobin		no
creatinine_phosphokinase	Feature	Integer		level of the CPK enzyme in the blood	mcg/L	no
diabetes	Feature	Binary		if the patient has diabetes		no
ejection_fraction	Feature	Integer		percentage of blood leaving the heart at each contraction	%	no
high_blood_pressure	Feature	Binary		if the patient has hypertension		no



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Variable Name	Role	Type	Demographic	Description	Units	Missing V
serum_creatinine	Feature	Continuous		level of serum creatinine in the blood	mg/dL	no
serum_sodium	Feature	Integer		level of serum sodium in the blood	mEq/L	no
sex	Feature	Binary	Sex	woman or man		no

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Additional Variable Information

Thirteen (13) clinical features:

- age: age of the patient (years)
- anaemia: decrease of red blood cells or hemoglobin (boolean)
- creatinine phosphokinase (CPK): level of the CPK enzyme in the blood (mcg/L)
- diabetes: if the patient has diabetes (boolean)
- ejection fraction: percentage of blood leaving the heart at each contraction (percentage)
- high blood pressure: if the patient has hypertension (boolean)
- platelets: platelets in the blood (kiloplatelets/mL)
- sex: woman or man (binary)
- serum creatinine: level of serum creatinine in the blood (mg/dL)
- serum sodium: level of serum sodium in the blood (mEq/L)
- smoking: if the patient smokes or not (boolean)
- time: follow-up period (days)
- [target] death event: if the patient died during the follow-up period (boolean)

For more information, please check Table 1, Table 2, and Table 3 of the following paper:

Davide Chicco, Giuseppe Jurman: "Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone". BMC Medical Informatics and Decision Making 20, 16 (2020).
https://doi.org/10.1186/s12911-020-1023-5

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Dataset Files

File	Size
heart_failure_clinical_records_dataset.csv	12 KB